

3D Visualization of Biological Data in Ultra High Definition Virtual Reality

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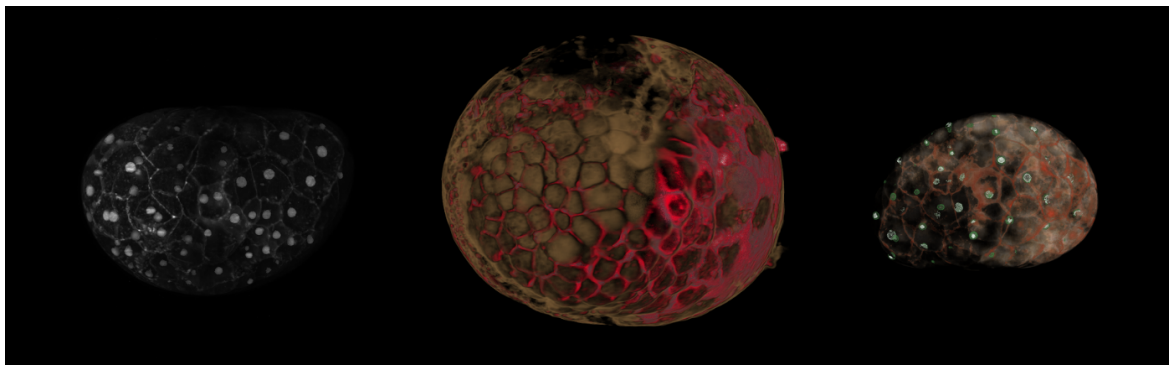


Figure 1: 3D reconstructed biological data

ABSTRACT

Visual computing is an indispensable tool in biology, revolutionizing the way researchers interact with complex biological data. The integration of high-resolution Virtual Reality offers rich visualization experience, crucial for advancing our knowledge in areas such as genomics, proteomics, and cellular biology. Recent VR-based biological visualization often deploys lower-resolution Head Mounted Displays (HMDs), limiting the observation of fine structural detail. Moreover, the simultaneous exploration of multiple datasets in an interactive manner is still a research challenge. This work presents an interactive 3D reconstruction and visualization system for biological data in ultra-high-definition VR. The system supports volume rendering, flexible rendering, and the simultaneous visualization of multiple datasets, all within an 8K VR environment. Intricate cellular and subcellular structures as well as detailed protein folding patterns and subtle variations in tissue density are visualized, critical for accurate medical diagnosis and research. Our system's performance is evaluated based on loading time, frames per second, and device latency, while handling large biological datasets. A think aloud study provided expert feedback.

Index Terms: Human-Centered Computing, Visualization, Visualization Techniques

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1 INTRODUCTION

Visual computing has become an indispensable tool in medicine and biology. [22, 17, 28]. As biological research focuses on the intricacies of cellular and molecular structures, current methods of data analysis and visualization often fall short in conveying intricate biological processes [15]. Visual computing transforms abstract data into intuitive, interactive visual representations [15]. This capability advances our knowledge in genomics, proteomics, and cellular biology. Thus, there is demand for novel visualization techniques and apparatus able to handle large and complex data [7].

Virtual Reality (VR) offers detailed examination of biological structures in 3D, facilitating deeper understanding of biological systems [30]. VR enables the visualization of high-resolution volumetric data, allowing for exploration of anatomical and molecular models with unprecedented clarity [8]. Tools such as VTK and ParaView [2] have been VR adapted, providing robust platforms for the visualization of large-scale biological datasets. The need for high-resolution Head Mounted Displays (HMDs) arises due to motion sickness caused by devices of lower resolution [22]. High-quality HMDs also eliminate artifacts present at lower resolution displays. Moreover, low resolution HMDs are not well-accepted clinically, due to their limited optical capabilities [15].

This work puts forward an innovative VR system designed to interactively visualize biological data in high fidelity. The VR system processes stacked image files to create 3D objects that can be explored in VR. By leveraging volume rendering techniques, our system aims to provide users with a high fidelity, immersive experience, allowing for detailed examination of complex biological structures such as neurons, tissues, and cellular components [14, 16]. Once an object is loaded, users can move and rotate it for thorough observation. They can also edit, save, and load the object's appearance and color settings (transform function) to enhance visual details. The system allows the use of a cross-section plane,

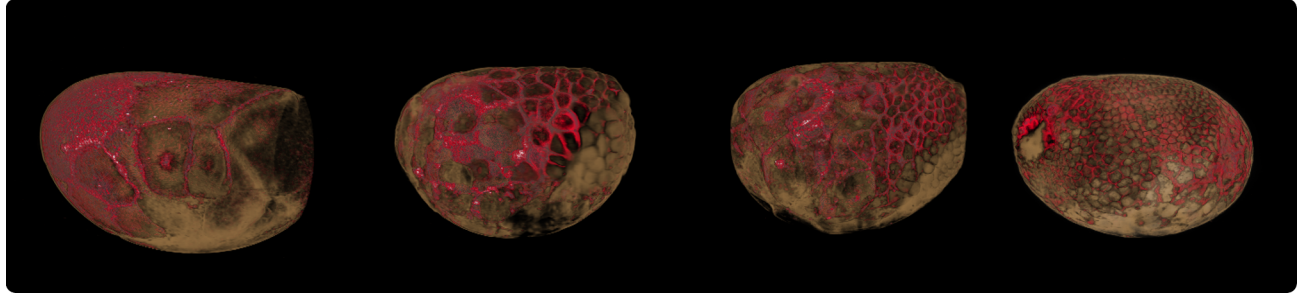


Figure 2: Results of 3d rendered object for different angles using Direct Volume Rendering technique

so as to analyze the inner parts of the object. Our contributions include:

- Our system loads multiple biological datasets simultaneously and fast, either all at once or one by one. Based on volume rendering processes, users seamlessly explore and interact with complex biological data from multi-view acquisitions based on 3D reconstruction, taken every 10 minutes or from the same acquisition captured in two separate channels.
- By using an 8K VR headset, we offer a level of visual detail of biological structures previously unattainable in VR, demonstrating the integration of cutting-edge VR technology with biological data visualization. The visualization as well as interaction with intricate cellular and subcellular structures are visualized as well as detailed protein folding patterns and subtle variations in tissue density.

2 RELATED WORK

Unity's volume rendering is often deployed to integrate volume rendering within the Unity environment [5, 19, 6], developed by Matias Lavik. This includes a wide set of tools for editing and manipulating reconstructed objects [23]. Our work builds upon Lavik's volume rendering system, extending its capabilities to VR.

Past work in VR-based biological visualization explored the use of 3D VR for preoperative planning in surgeries, providing surgeons with an intuitive understanding of patient-specific anatomy [26, 1]. ProteinVR was introduced as web-based, for stereoscopic 3D visualization of molecular structures [4]. A VR-based volumetric medical image segmentation and visualization system was developed, offering interaction with 3D volumes [10]. A VR-based model for visualizing gene-gene interactions was presented [27]. While past work highlights the potential of VR in biological visualization, it deploys HMDs and overall visualization of lower resolution which may fail to capture the fine detail of biological structures.

High-resolution visualization is crucial for examining fine cellular and sub-cellular structures, such as the detailed morphology of neurons [12], the complex folding patterns of proteins [13], and the subtle variations in tissue density [21] that are critical for accurate diagnosis and research. The ability to visualize these intricate details in higher resolution significantly enhances both research and clinical applications, providing precise data interpretation [18]. In this work, we leverage 8K VR, offering significantly higher visual fidelity and detail, allowing users to examine complex biological datasets, in real-time, with greater clarity.

Past work reviewed high-resolution 3D tractography of fibrous tissue based on polarization-sensitive optical coherence tomography [32]. This technique provided detailed visualization of fibrous structures in biological tissues on a desktop platform, crucial for understanding tissue organization and pathology. This work did not

support the simultaneous exploration of multiple datasets in immersive VR. Our work allows users to open and view multiple objects, either as different channels of the same object or multiple instances of the same object from different time points. This is useful for comparative and temporal analysis, as users can switch between datasets fast, as they are pre-loaded in the system.

3 DATA ACQUISITION

Biological data were generated from time-lapse microscopy recordings of live developing embryos from the crustacean model organism *Parhyale hawaiiensis* [24]. *Parhyale* embryos were fluorescently labelled with the membrane-tethered marker Lyn-mCherry marking the outlines of all cells (Figure 2). Imaging was performed on an adaptive SiMView light-sheet fluorescence microscope with four orthogonal optical arms, which simultaneously illuminated the embryo with two 594 nm laser light-sheets from two opposite sides and acquired serial optical sections from the two perpendicular sides with water-dipping 16x/0.8 detection objectives [31, 25]. Multi-view acquisitions of live *Parhyale* embryos were carried out every 10 min at 26°C, over a period of 60 hrs covering the early cleavage, gastrulation and germband formation stages (Figure 2). TIFF files of acquired raw views (z-stacks) were first corrected and resaved in lossless compression, block-based file formats (HDF5 or KLB), were then spatiotemporally registered and fused into a single output 3D image volume per time point using established MATLAB and Fiji/ImageJ pipelines [3, 11].

4 VISUALIZATION OF BIOLOGICAL DATA IN VR

4.1 System Overview

Our VR system for 3D visualization of biological data is structured around four cohesive, modular key pillars: pre-processing, 3D visualization, VR, tools and features.

4.2 Pre-Processing

In the pre-processing stage, the original data was provided in HDF5 format [9], which, while useful for storing complex data structures and large datasets, it needed to be unpacked for use in our system. The selected format was NRRD (Nearly Raw Raster Data), well-suited for volumetric data used in biological visualization. A Python script was developed that reads the HDF5 files and generates the datasets in NRRD format, including varied resolutions for each dataset to ensure efficient processing and rendering. This script automates the conversion process, ensuring that the data is formatted and optimized for subsequent stages of the application.

4.3 3D Visualization

After the files are in the NRRD format, the next step is to load them into Unity for visualization. When a file is opened, Unity reads the file through an importer script that uses SimpleITK, a simplified layer built on top of the Insight Segmentation and Registration

Toolkit (ITK). SimpleITK facilitates the reading and processing of medical images, making it an ideal choice for handling our biological data. Once the raw data are read, they are stored in a 3D texture. This texture serves as the basis for creating a 3D model of the biological structures. The core technique employed for rendering this 3D model is raymarching, a method that traces rays through the 3D texture to compute densities and generate a volumetric representation of the data. Raymarching is particularly effective for visualizing complex volumetric data, as it allows for the detailed and accurate rendering of internal structures.

After the 3D model is generated, there are two options for coloring the object implemented: Maximum Intensity Projection (MIP) and Direct Volume Rendering (DVR). Maximum Intensity Projection works by projecting the highest intensity value encountered along each ray onto the viewing plane, which highlights the brightest structures within the volume, making it useful for emphasizing features such as bone in medical imaging [29]. DVR involves computing color and opacity values along each ray and integrating these values to produce the final image (Figure 5(a)), allowing for a more comprehensive representation of the data (Figure 2) employing depth and shading that can highlight subtle variations within the structures [20]. Both techniques offer unique advantages, and the choice between them depends on the specific visualization needs of the user.

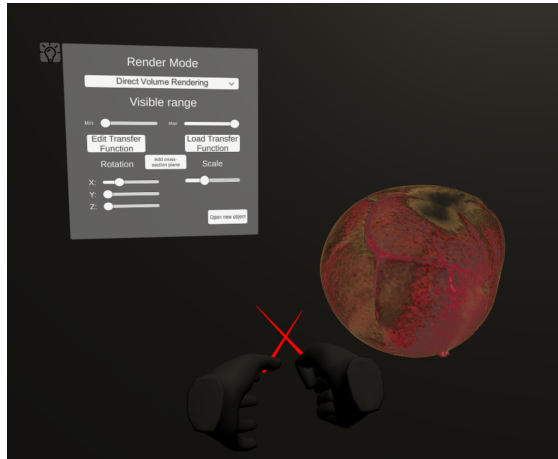


Figure 3: The virtual environment

4.4 VR Visualization

After the visualization is complete, the next step is to create the virtual scene. Initially, we convert the single camera setup into a dual-camera system, enabling stereoscopic rendering necessary for visualization in VR. Next, we deploy the high resolution Pimax 8K VR headset (Figure 4). The connection of the headset to the PC is managed by PiTool, a proprietary software from Pimax. For integrating our system with the headset, we utilized SteamVR, which supports various VR headsets. We implemented the OpenXR standard, which not only optimizes performance but also ensures compatibility with a wide range of VR headsets beyond just the Pimax 8K, enhancing compliance by various systems (Figure 3)

Users can access the user interface (UI) of our VR system with the press of a button located on their hand controllers. Users navigate the VR system and access the necessary tools without interrupting their immersive experience. The interface allows for toggling the environment light to increase contrast during the observation phase. This feature helps to highlight details within the biological models while keeping the light on during other times to maintain spatial orientation and reduce the likelihood of motion sickness.

Users can move and rotate objects using the controllers, allowing for an interactive examination of the 3D models. For more precise adjustments, sliders are available to fine-tune rotations. Users can change the scale of the objects, zooming in on specific areas for a closer look, in 8K high fidelity. A detailed visualization together with intuitive manipulation tools enhances user experience and analysis of complex biological structures.



Figure 4: Use of Pimax and Valve index controllers

4.5 Tools and Features

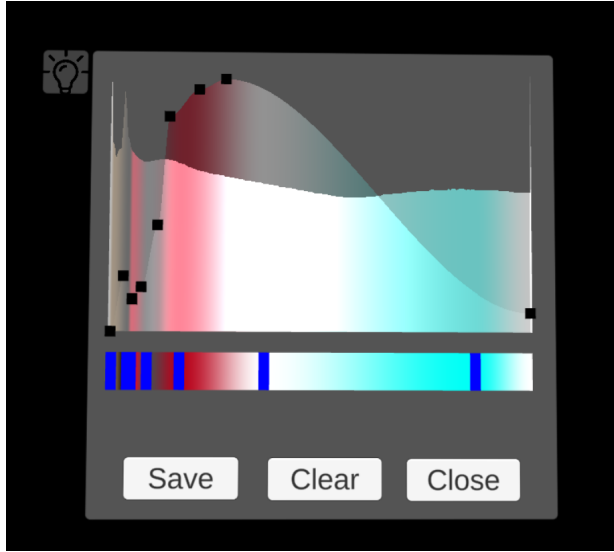
When the rendering mode is set to DVR instead of MIP, users manipulate the color and opacity of each density value, through a transfer function editor that displays a histogram of the densities as shown in Figure 5(a). Users add and move color and alpha knots on this histogram, for fast, real-time adjustments to the appearance of the objects, fine-tuning the visualization quickly and efficiently. Users save their customized transfer functions or load pre-existing ones, avoiding the need to redo adjustments each time a dataset is reopened. Users may also add cross-plane sections, cutting through the 3D object observing internal structures, shown in Figure 5(b). This feature is useful for examining the interior of complex biological models, providing a clear and detailed view of data from different angles and sections.

Our system also supports opening of multiple files simultaneously, as: a single file, time instances, and channels. In single file mode, a single object is reconstructed and displayed. The time instances mode allows for multiple objects to be loaded, with only one displayed at a time. A slider is provided to switch between different time points, enabling users to observe changes in density and form over time while using the same transfer function for consistency. In channels' mode, multiple objects are loaded and displayed simultaneously, which is useful when different parts of a structure are captured separately but need to be studied together, such as membranes and cores. In this mode, each object (or channel) has its own transfer function and can be edited separately, allowing for detailed comparative analysis and comprehensive visualization.

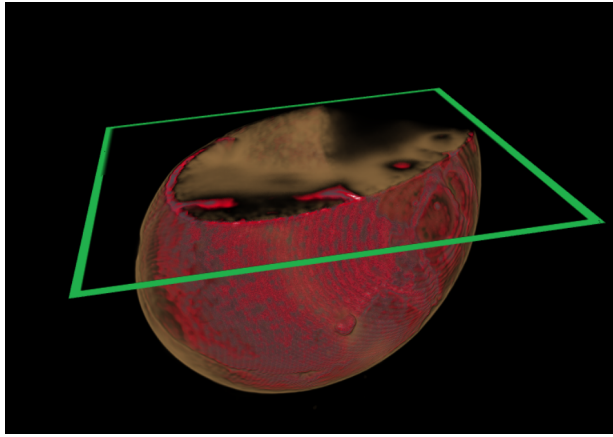
5 EVALUATION

The system used for the development and evaluation was equipped with an Intel i7 processor, NVIDIA RTX 3070 GPU, 64 GB of DDR4 RAM, and ran on Windows 11.

The evaluation of our VR system was conducted to measure its performance using loading time, frames per second (FPS), and de-



(a) Editor window of the transfer function



(b) Cross Plane on a reconstructed object

Figure 5: Tools and functionalities

vice latency metrics, assessing system responsiveness while handling large biological datasets in VR. We collected measurements for each metric across five resolutions of a dataset. For each resolution, ten measurements were acquired for loading time, FPS, and latency. The average value was then calculated to provide a representative performance metric for each resolution.

Loading time, demonstrated a nearly linear increase as the size of the dataset grew which is expected as the loading process is CPU-bound. In relation to FPS, the VR system consistently managed to run at an average of around 100 FPS for all resolutions, except the highest one. For that, the FPS dropped significantly, indicating that our system exceeded the capabilities of our graphics card (Nvidia RTX 3070). GPU capacity is shown to be the limiting factor for fast interaction. The Pimax 8K VR headset maintained consistent latency of under 7 ms across all tests, except for the highest resolution dataset, for which it was 100 ms, attributed to the system's lower performance. System optimization is crucial for maintaining smooth and responsive interaction, for ultra high-resolution data.

10 VR experts based on a research laboratory (mean age 26 years old, 6 male, 4 female) employed a think aloud methodology to eval-

	X	Y	Z
Lowest	140	131	108
Low	280	262	217
hline	561	525	434
Medium			
High	1122	1050	868
Highest	2244	2100	1736

Table 1: Dimension sizes for each dataset evaluated

	Loading Time (ms)	FPS	Latency (ms)
Lowest	34	109	5.32
Low	161	106	5.33
Medium	1238	100	4.84
High	9970	100	5.42
Highest	81519	2	105.5

Table 2: Evaluation results

uate the usability and functionality of our VR system. Participants were tasked with loading two distinct datasets and edit their transform functions as well as change their appearance to maximize the observable details. They verbalized their thoughts, actions, and observations, providing real-time insights into their interaction with the app. Feedback was categorized into comments about the hardware (HMD and controllers) and the VR system itself. Participants noted that the HMD felt heavy after extended use, and the grab action on the controllers did not always register correctly. They appreciated the external light cut-off as the HMD fit well to the face. Participants reported that the precision of interactions via the controllers' rays decreased as the distance to the object or UI increased. When objects overlapped, such as in multi-instance open mode or when a cross-section plane overlapped with another object, it was unclear which object would be grabbed with the grab action. While adjusting the transform function, users found it challenging to accurately place nodes as they focused on the real-time changes in the object and not on the placement of the node. Participants did not report any sense of motion sickness.

6 CONCLUSIONS AND FUTURE WORK

We developed a novel, interactive 3D visualization system of biological data in ultra high-definition VR, employing an 8K VR headset. The VR system handles large datasets, providing high-fidelity visualization through volume rendering. It loads multiple biological datasets simultaneously and fast, from multi-view acquisitions based on 3D reconstruction, taken every 10 minutes or from the same acquisition captured in two separate channels. High-resolution HMDs enable the visualization of intricate cellular and subcellular structures, detailed protein folding patterns, and subtle variations in tissue density, all of which are critical for accurate medical diagnosis and research. This paper validates the feasibility of implementing high-resolution, interactive, VR systems for biological 3D visualization with the potential to revolutionize the way researchers and practitioners interact with complex biological data.

Future work includes the addition of sophisticated tools for data manipulation, such as advanced annotation tools for marking specific regions of interest, and exporting tools in order to save screenshots and measurements that have been acquired during analysis. Improved support for collaborative VR environments would allow multiple users to interact with the same dataset in real time, facilitating collaborative research and decision-making processes in medical and biological studies. The integration of machine learning algorithms may assist in data analysis by automating the identi-

fication of patterns and anomalies within large datasets, thereby enhancing accuracy and efficiency in research. Leveraging advancements in hardware, such as more powerful GPUs, faster processors as well as more optimized rendering algorithms and VR frameworks, will be crucial for improving performance, and, thus, enhancing the user experience. In order to evaluate the applicability of the system in the biological analysis and research context, a future formal analysis with an extended target audience (medical, biomedical, and biological practitioners and researchers) is necessary. This will provide valuable feedback related the system's usability, functionality, and overall impact, ensuring that it meets the specific needs and expectations of its intended users.

ACKNOWLEDGMENTS

- This work has been supported by the Horizon Europe Research & Innovation Programme under Grant agreement N. 101092612 (Social and hUman ceNtered XR - SUN project). Views and opinions expressed in this work are those of the author(s) only and do not necessarily reflect those of the European Union. Neither the European Union nor the European Commission can be held responsible for them.

REFERENCES

- [1] D. Abjigitova, A. H. Sadeghi, J. J. Peek, J. A. Bekkers, A. J. J. C. Bogers, and E. A. F. Mahtab. Virtual reality in the preoperative planning of adult aortic surgery: A feasibility study. *Journal of Cardiovascular Development and Disease*, 9(2), 2022. doi: 10.3390/jcdd9020031 2
- [2] J. Ahrens, B. Geveci, and C. Law. Paraview: An end-user tool for large data visualization. *Visualization Handbook*, 01 2005. 1
- [3] F. Amat, B. Höckendorf, Y. Wan, W. C. Lemon, K. McDole, and P. J. Keller. Efficient processing and analysis of large-scale light-sheet microscopy data. *Nature Protocols*, 10(11):1679–1696, Nov 2015. doi: 10.1038/nprot.2015.111 2
- [4] K. Cassidy, J. Sefcik, Y. Raghav, A. Chang, and J. Durrant. Proteinvr: Web-based molecular visualization in virtual reality. *PLOS Computational Biology*, 16:e1007747, 03 2020. doi: 10.1371/journal.pcbi.1007747 2
- [5] B. Cetinsaya, C. Neumann, and D. Reinert. Using direct volume rendering for augmented reality in resource-constrained platforms. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 768–769, 2022. doi: 10.1109/VRW55335.2022.00235 2
- [6] S. Cetkowski. Virtual reality visualization of computed tomography for product development process. 2020. 2
- [7] M. Chen, D. Ebert, H. Hagen, R. S. Laramée, R. van Liere, K.-L. Ma, W. Ribarsky, G. Scheuermann, and D. Silver. Data, information, and knowledge in visualization. *IEEE Computer Graphics and Applications*, 29(1):12–19, 2009. doi: 10.1109/MCG.2009.6 1
- [8] K. Engel, M. Kraus, and T. Ertl. High-quality pre-integrated volume rendering using hardware-accelerated pixel shading. *Proceedings of the ACM SIGGRAPH Conference on Computer Graphics*, 06 2001. doi: 10.1145/383507.383515 1
- [9] M. Folk, G. Heber, Q. Koziol, E. Pourmal, and D. Robinson. An overview of the hdf5 technology suite and its applications. pp. 36–47, 03 2011. doi: 10.1145/1966895.1966900 2
- [10] Y. Gao, C. Chang, X. Yu, P. Pang, N. Xiong, and C. Huang. A vr-based volumetric medical image segmentation and visualization system with natural human interaction. *Virtual Reality*, 26(2):415–424, Jun 2022. doi: 10.1007/s10055-021-00577-4 2
- [11] D. Hörl, F. Rojas Rusak, F. Preusser, P. Tillberg, N. Randel, R. K. Chhetri, A. Cardona, P. J. Keller, H. Harz, H. Leonhardt, M. Treier, and S. Preibisch. Bigstitcher: reconstructing high-resolution image datasets of cleared and expanded samples. *Nature Methods*, 16(9):870–874, Sep 2019. doi: 10.1038/s41592-019-0501-0 2
- [12] I. Izeddin, C. G. Specht, M. Lelek, X. Darzacq, A. Triller, C. Zimmer, and M. Dahan. Super-resolution dynamic imaging of dendritic spines using a low-affinity photoconvertible actin probe. *PLOS ONE*, 6(1):1–14, 01 2011. doi: 10.1371/journal.pone.0015611 2
- [13] J. Juárez-Jiménez, P. Tew, M. O'Connor, S. Llabrés, R. Sage, D. Glowacki, and J. Michel. Combining virtual reality visualization with ensemble molecular dynamics to study complex protein conformational changes. *Journal of Chemical Information and Modeling*, 60(12):6344–6354, 2020. doi: 10.1021/acs.jcim.0c00221 2
- [14] A. Kaufman and K. Mueller. *Overview of Volume Rendering*, vol. 7, pp. 127–X1. 12 2005. doi: 10.1016/B978-012387582-2/50009-5 1
- [15] J. Klein, O. Friman, M. Hadwiger, B. Preim, F. Ritter, A. Vilanova, G. Zachmann, and D. Bartz. Visual computing for medical diagnosis and treatment. *Computers and Graphics*, 33(4):554–565, 2009. doi: 10.1016/j.cag.2009.04.006 1
- [16] J. Kniss, G. Kindlmann, and C. Hansen. Interactive volume rendering using multi-dimensional transfer functions and direct manipulation widgets. 01 2002. doi: 10.1109/VISUAL.2001.964519 1
- [17] G. Kontidakis, D. Chasiouras, D. Proimaki, M. Halkiadakis, M. Fyntikaki, and K. Mania. Gamified platform for rehabilitation after total knee replacement surgery employing low cost and portable inertial measurement sensor node. *Multimedia Tools and Applications*, 79, 02 2020. doi: 10.1007/s11042-018-6572-6 1
- [18] G. Koulieris, K. Akşit, M. Stengel, R. Mantiuk, K. Mania, and C. Richardt. Near-eye display and tracking technologies for virtual and augmented reality. *Computer Graphics Forum*, 38:493–519, 06 2019. doi: 10.1111/cgf.13654 2
- [19] M. Lavik. Unity volume rendering. 2019. 2
- [20] M. Levoy. Display of surfaces from volume data. *IEEE Computer Graphics and Applications*, 8(3):29–37, 1988. doi: 10.1109/38.511 3
- [21] H. Liu, C. Zhang, X. Fan, Y. Duan, T. Xiao, G. Du, Y. Fu, H. Liu, and H. Wen. Robust phase-retrieval-based x-ray tomography for morphological assessment of early hepatic echinococcosis infection in rats. *PLOS ONE*, 12(9):1–14, 09 2017. doi: 10.1371/journal.pone.0183396 2
- [22] A. J. Olson. Perspectives on structural molecular biology visualization: From past to present. *Journal of Molecular Biology*, 430(21):3997–4012, 2018. doi: 10.1016/j.jmb.2018.07.009 1
- [23] E. Prodromou, S. Leandrou, E. Schiza, K. Neocleous, M. Matsangidou, and C. Pattichis. A multi-user virtual reality application for visualization and analysis in medical imaging. pp. 795–800, 2020. doi: 10.1109/BIBE50027.2020.00135 2
- [24] J. Rallis, G. Kapai, and A. Pavlopoulos. *Handbook of Marine Model Organisms in Experimental Biology: Established and Emerging*, chap. 16. CRC Press, 2021. 2
- [25] L. A. Royer, W. C. Lemon, R. K. Chhetri, Y. Wan, M. Coleman, E. W. Myers, and P. J. Keller. Adaptive light-sheet microscopy for long-term, high-resolution imaging in living organisms. *Nature Biotechnology*, 34(12):1267–1278, Dec 2016. doi: 10.1038/nbt.3708 2
- [26] A. Sadeghi, W. Bakhuis, F. Van Schaagen, F. Oei, J. Bekkers, A. Maat, E. Mahtab, A. Bogers, and Y. Taverne. Immersive 3d virtual reality imaging in planning minimally invasive and complex adult cardiac surgery. *European heart journal*, p. 62–70, 2020. doi: 10.1093/ehjdh/ztaa011 2
- [27] A. Saleh and B. Mahmood. Vr-based visualization for biological networks. 7:48–68, 05 2023. doi: 10.47001/IRJIET/2023.705007 2
- [28] F. Sarri, P. Kasnesis, S. Symeonidis, I. Th, I. Paraskevopoulos, S. Diplaris, F. Posteraro, G. Georgoudis, and K. Mania. Embodied augmented reality for lower limb rehabilitation. 09 2024. 1
- [29] Y. Sun and D. Parker. Performance analysis of maximum intensity projection algorithm for display of mra images. *IEEE Transactions on Medical Imaging*, 18(12):1154–1169, 1999. doi: 10.1109/42.819325 3
- [30] R. P. Theart, B. Loos, and T. R. Niesler. Virtual reality assisted microscopy data visualization and colocalization analysis. *BMC Bioinformatics*, 18(2):64, Feb 2017. doi: 10.1186/s12859-016-1446-2 1
- [31] R. Tomer, K. Khairy, F. Amat, and P. J. Keller. Quantitative high-speed imaging of entire developing embryos with simultaneous multiview light-sheet microscopy. *Nature Methods*, 9(7):755–763, Jul 2012. doi: 10.1038/nmeth.2062 2
- [32] G. Yao and D. Duan. High-resolution 3d tractography of fibrous tissue based on polarization-sensitive optical coherence tomography. *Experimental Biology and Medicine*, 245(4):273–281, 2020. PMID: 31813275. doi: 10.1177/1535370219894332 2